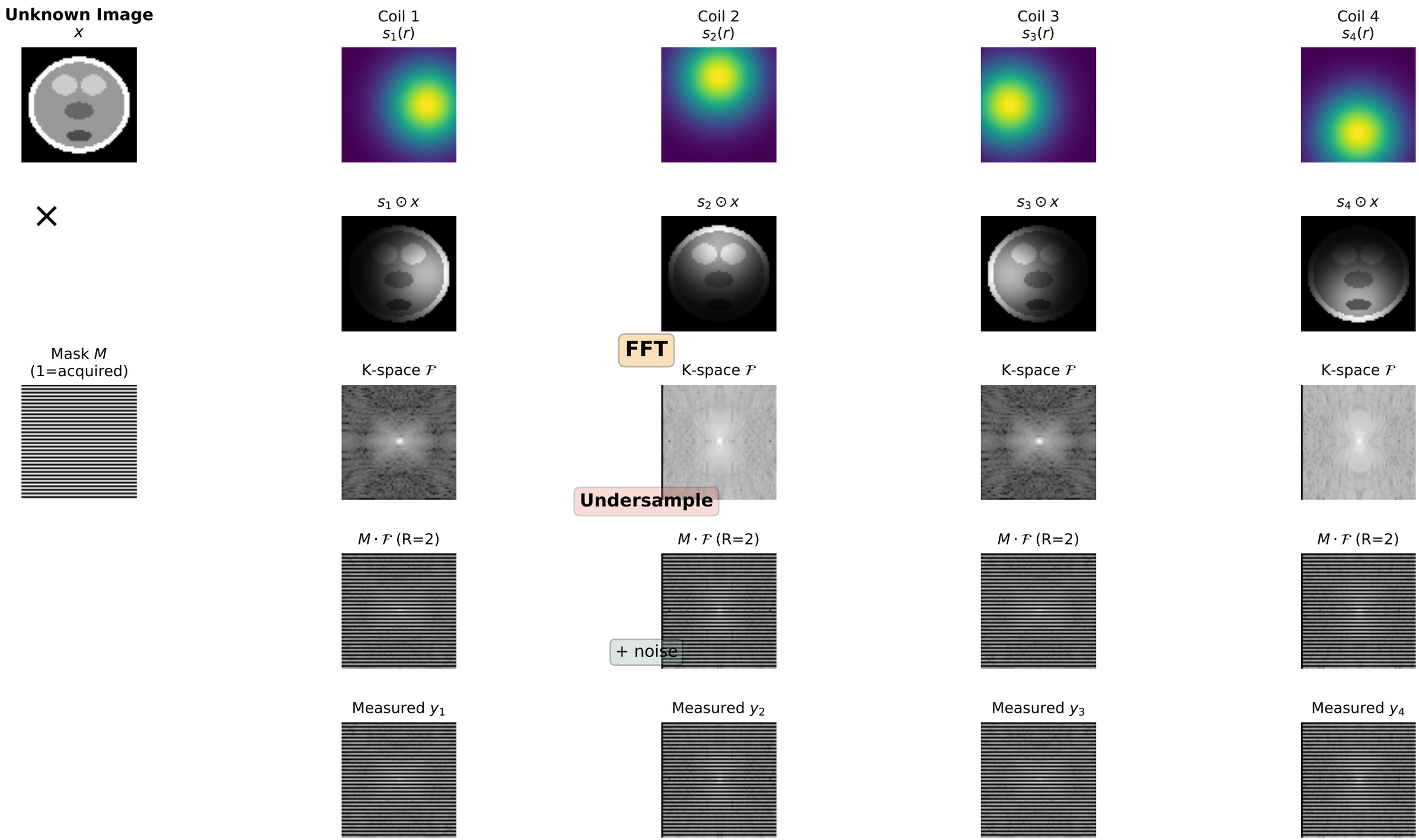
# Multi-Coil MRI Forward Model: $y_i = M\mathcal{F}\{(S_i \odot x)\} + \eta_i$




**SENSE Inverse Problem**

$$\min_x \sum_{i=1}^{C} \|M\mathcal{F}S_i x - y_i\|_2^2$$

*Solve with Conjugate Gradient*